



Autonomous Detection of Pneumonia in Toddlers using a Self-Assembled Digital Stethoscope

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A project report submitted in partial fulfillment of the requirements for the degree
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May 2023

DECLARATION

This Final Year Capstone Project report is submitted to the department of Electrical and Electronic Engineering, University of Liberal Arts Bangladesh in partial fulfillment of the requirements for the degree of Bachelor of Science. So, we hereby declare that this report is based on the work done by us. Materials of work found by other researchers are mentioned by reference. This project report, neither in whole nor in part, has been previously submitted for any degree.

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ACKNOWLEDGEMENT

First and foremost, we would like to thank Almighty Allah for giving us the knowledge, health, and patience needed to finish this Final Year Capstone Project.

We also want to thank Dr. Mirza Rasheduzzaman, an Associate Professor of Electrical and Electronic Engineering and Dr.A.B.M Sayeed Ud Doulah, an Assistant Professor and Coordinator of Electrical and Electronic Engineering at the University of Liberal Arts Bangladesh (ULAB), for setting up the completion of this Final Year Capstone Project for us as well as for his excellent guidance, encouragement, and kindness.

Our department esteemed all the faculty members, in particular Professor Dr. M. Mofazzel Hossain, Dr. Litton Hossain, MD. Shahjalal, Ramit Kumar Sadhukhan, and Atik Jawad are to be thanked for their guidance, co-operation and lectures during the study.

We would also like to thank Mr. Khorshed Alam and Mr. Shamim our gratitude to the Assistant Officer, for his four years of cooperation at the university.

We also want to express my gratitude to our family and friends, who supported and encouraged us emotionally during the endeavor. Our motivation and focus have been maintained in large part by their unfailing support.

Finally, we would like to thank everyone who took part in the research and gave us their insightful opinions and input, without whose help this project would not have been feasible.

We appreciate your help and support with this initiative.

ABSTRACT

This project presents the development of an innovative and comprehensive digital stethoscope system aimed at enhancing the detection and diagnosis of pneumonia in pediatric patients. The system integrates advanced technologies, including high-fidelity audio recording, audio files are preprocessed using Mel-frequency cepstral coefficients (MFCCs), feature and machine learning algorithms (Convolutional Neural Networks (CNNs), to provide accurate and real-time analysis of lung sounds for improved diagnostic capabilities. The study has used several subjects (toddlers) and there has been a total of 10 pneumonia patient kids and 9 Healthy kids during this project. The digital stethoscope system allows healthcare professionals to capture and transmit lung sound data from pediatric patients to a computer or mobile device. This data is then processed using sophisticated signal processing techniques and machine learning algorithms, enabling the identification of abnormal breath sounds associated with pneumonia. The achieved overall accuracy of 76% indicates that the model can differentiate between healthy and pneumonia-affected toddlers to a reasonable extent. By the system can effectively distinguish between normal and pathological lung sounds, facilitating early detection and timely intervention. In addition to its diagnostic capabilities, there are some limitations as well. However, it is important to consider the limitations of the study, such as the sample size and potential biases in the dataset. Future research should focus on expanding the dataset, addressing potential biases, and conducting further validation studies. Additionally, efforts should be made to integrate the developed model into real-world clinical settings and evaluate its performance in practical scenarios. The affordability, portability, and ease of use of the digital stethoscope system make it particularly well-suited for deployment in resource-limited settings. By providing healthcare professionals with a reliable and efficient tool for pneumonia diagnosis, the system has the potential to significantly impact pediatric healthcare outcomes, especially in regions where access to specialized medical equipment and expertise may be limited. Overall, this project represents a significant advancement in the field of pediatric pneumonia diagnosis, offering a comprehensive digital stethoscope system that combines cutting-edge technologies, machine learning algorithms. The system's capabilities in early detection, accurate diagnosis, and seamless data management have the potential to improve patient outcomes, reduce healthcare costs, and enhance the overall quality of care for pediatric patients with pneumonia.

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List of Abbreviation

Abbreviation	Full Form
PCG	Phonocardiogram
ECG	Electrocardiogram
S1	First heart sound
S2	Second heart sound
TK	Taka (Bangladeshi currency)
HL7	Health Level 7 (standard for exchanging health data)
NHI	National Health Identification
Mirth	Mirth Connect (healthcare integration engine)
CNN	Convolutional Neural Network
MFCC	Mel-frequency Cepstral Coefficients
API	Application Programming Interface
GUI	Graphical User Interface

CHAPTER ONE

INTRODUCTION

1.1 Background

Pneumonia, the leading infectious cause of death in children under the age of five worldwide [1], poses a significant health burden, accounting for an estimated 800,000 fatalities annually, with over half occurring in just five low- and middle-income countries (LMICs) [1, 2]. In developing countries, the projected yearly incidence of pneumonia in children under five years old was approximately 231 episodes per 1,000 children in 2015, resulting in a staggering 138 million clinical pneumonia cases within this age group [3]. Timely recognition of symptoms and prompt treatment are crucial for reducing child mortality associated with pneumonia [4].

Pneumonia is a common and potentially life-threatening respiratory infection, particularly affecting toddlers. Early detection and treatment play a vital role in improving outcomes and reducing mortality rates. However, diagnosing pneumonia in children can be challenging, especially in resource-limited settings where access to medical resources and knowledge may be constrained.

Recent advancements in technology and artificial intelligence have opened new possibilities for transforming the diagnostic process of pediatric pneumonia [1]. Among these developments, digital stethoscopes have emerged as promising instruments for the identification of respiratory disorders [5]. Unlike traditional stethoscopes, digital variants possess the ability to record and analyze lung sounds, enabling healthcare professionals to make more accurate and timely diagnosis [5]. In particular, the autonomous detection of pneumonia in children using self-assembled digital stethoscopes has garnered significant attention within the research community [6].

By harnessing the power of machine learning algorithms, these self-assembled digital stethoscopes can swiftly and precisely diagnose pneumonia by detecting specific sound patterns associated with the diseases. This innovative approach offers numerous advantages over conventional methods. It reduces the dependence on medical experts, making it especially valuable in settings with limited healthcare resources [7]. Additionally, the affordability and non-invasiveness of digital stethoscopes make them a viable alternative to established diagnostic techniques like chest X-rays, potentially serving as a complementary tool in the diagnostic process [8].

Traditional stethoscopes have inherent limitations that hinder their effectiveness in capturing high-frequency sounds such as wheezing and crackles [9, 10]. Furthermore, they tend to transmit ambient noises and tube resonance, impeding accurate auscultation [9, 10]. The advent of real-time lung sound categorization or digital auscultation has addressed these challenges [11]. By leveraging artificial intelligence algorithms, digital auscultation exhibits the potential to become a highly specific diagnostic tool for respiratory conditions, particularly in lower middle-income countries (LMICs) where healthcare resources are often scarce [12]. Introducing adventitious sound categories into Integrated Management of Childhood Illness (IMCI) recommendations could potentially minimize the unnecessary administration of antibiotics in LMICs, further enhancing the diagnostic accuracy and appropriateness of treatment [12].

Conventional auscultation with an analog stethoscope requires a quiet environment and a cooperative patient, which is troublesome in hospitals/clinics, especially in LMICs where patient numbers often exceed capacity. Exam rooms are generally full with chatting individuals, ringing phones, and whirling fans; the

child being examined may be nervous, rebellious, or crying. To improve hearing, digital stethoscopes decrease ambient noise [12, 13]. Conventional auscultation with a stethoscope has limits owing to the human auditory system, which ranges from 20 Hz to 20 kHz in youth and decreases beyond middle age. Digital stethoscopes may magnify sounds 100-fold [14]. When interpreting body sounds, inter-rater variability is high, regardless of the amount of expertise of the practitioner. Modern advances in ML and AI have allowed the efficient analysis of complicated data, such as lung sounds and rhythms. In [15], Machine Learning achieved >90% illness diagnostic accuracy. This research [15-17] uses machine learning and AI to detect lung disorders.

In a previous work conducted by three researchers [7], the objective was to enhance conventional acoustic stethoscopes by integrating a transducer, specifically a condenser microphone, into the tube. The digital signals captured by the transducer were stored and processed to extract relevant features. Using the Pearson correlation coefficient, the extracted features were evaluated and compared. The results indicated that an MDF Acoustic Stethoscope, which was readily available in the local market, exhibited the potential to capture high-quality data, laying the foundation for the development of a machine learning algorithm.

Building upon this prior research, the present study focuses on the collection of high-quality lung sounds from both healthy and unhealthy subjects, specifically infants and toddlers. The primary aim is to compile a comprehensive database of lung sounds that encompasses a range of respiratory conditions. This extensive dataset will serve as the basis for training and implementing a machine learning algorithm capable of predicting the occurrence of pneumonia. By utilizing advanced computational techniques, the algorithm will leverage the distinctive features extracted from the collected lung sounds to accurately classify and identify pneumonia cases.

1.2 Problem Statement

Accurate and timely diagnosis of respiratory diseases, such as pneumonia, bronchitis, and chronic obstructive pulmonary disease (COPD), remains a significant challenge in the field of healthcare. Conventional auscultation methods that rely on acoustic stethoscopes are subjective and prone to interpretation variations, leading to potential errors and inconsistencies in diagnosis. Moreover, limited access to specialized healthcare facilities and trained medical professionals in remote or resource-limited areas further exacerbates the difficulty of detecting respiratory conditions promptly and accurately.

To address these pressing concerns, there is a critical need for an innovative and cost-effective solution that can improve the accuracy and accessibility of respiratory disease diagnosis. This solution should leverage technological advancements, including digital stethoscopes and artificial intelligence (AI), to provide reliable and objective assessments of lung sounds for precise detection and classification of respiratory conditions.

The problem at hand encompasses the following challenges:

Subjectivity and interpretation variability in auscultation: Traditional auscultation methods heavily rely on the experience and expertise of healthcare professionals, introducing subjectivity and inconsistencies in the interpretation of lung sounds. Objective and standardized approaches are necessary to enhance accuracy and consistency in respiratory disease diagnosis.

Limited access to specialized healthcare facilities and trained professionals: Numerous regions, particularly remote and underserved areas, face challenges in accessing specialized healthcare facilities and skilled medical professionals. This limited access results in delayed diagnoses, inadequate treatment, and compromised patient outcomes.

Affordability and portability of diagnostic tools: Conventional diagnostic equipment for respiratory conditions tends to be expensive and impractical for widespread use, especially in resource-constrained settings. A cost-effective and portable diagnostic solution is required, allowing for easy deployment and utilization by healthcare providers in diverse healthcare settings.

Integration of technology for precise diagnosis: Incorporating technological advancements, such as digital stethoscope technology and AI algorithms, can significantly enhance the accuracy and efficiency of respiratory disease diagnosis. The proposed solution should integrate machine learning techniques to analyze lung sounds and provide healthcare professionals with objective diagnostic information.

By addressing these challenges, the proposed solution aims to enhance the accuracy, accessibility, and affordability of respiratory disease diagnosis. This, in turn, has the potential to facilitate early intervention, improve patient outcomes, and bridge the healthcare access gap between different regions, ultimately contributing to enhanced respiratory health management on a global scale.

1.3 Motivation

The motivation to explore the utilization of digital stethoscopes and AI algorithms for respiratory condition diagnosis and monitoring, particularly pneumonia, arises from several crucial factors.

Pneumonia remains a significant global health concern, causing substantial morbidity and mortality, especially among vulnerable populations such as children and the elderly. Despite advancements in medical technology, accurately and promptly diagnosing pneumonia can be challenging, leading to treatment delays and potential adverse outcomes. By harnessing innovative technologies like digital stethoscopes and AI algorithms, there is an opportunity to enhance diagnostic accuracy and efficiency, ultimately resulting in improved patient outcomes.

The integration of AI and digital devices in healthcare has demonstrated tremendous potential across various domains. AI algorithms exhibit remarkable capabilities in pattern recognition, data analysis, and predictive modeling, which can be leveraged to analyze complex respiratory sounds and identify distinct patterns associated with pneumonia. Combining these algorithms with digital stethoscopes enables the creation of powerful diagnostic tools that empower healthcare professionals to make informed decisions, facilitating early detection and prompt treatment.

Moreover, these advanced diagnostic systems align with the growing trend of personalized and precision medicine. Each patient's condition is unique, and accurate diagnosis plays a pivotal role in tailoring treatment plans to individual needs. By harnessing the power of AI and digital stethoscopes, the precision and efficacy of respiratory disease diagnosis can be enhanced, enabling personalized treatment strategies that optimize patient outcomes.

Additionally, the integration of microcontrollers like Raspberry Pi offers numerous advantages, including cost-effectiveness, versatility, and connectivity. These microcontrollers provide platforms for real-time signal processing, data storage, and seamless integration with other healthcare systems. This opens possibilities for remote monitoring, telemedicine, and data-driven research, allowing healthcare professionals to extend their reach and deliver quality care to underserved populations, particularly in resource-limited settings.

Furthermore, advancements in technology have significantly reduced barriers to developing innovative healthcare solutions. Digital stethoscopes and microcontrollers are increasingly affordable and accessible, making it feasible to implement such systems in diverse healthcare settings. By leveraging these technologies

and fostering collaboration among multidisciplinary teams, progress in respiratory healthcare can be accelerated, contributing to global efforts in reducing the burden of pneumonia and other respiratory conditions.

In summary, the motivation to explore the utilization of digital stethoscopes, AI algorithms, and microcontrollers for respiratory disease diagnosis arises from the urgent need to enhance diagnostic accuracy, personalize treatment strategies, leverage the power of AI, and overcome barriers to accessing quality healthcare. By combining these technologies, there is a potential for revolutionizing respiratory healthcare, improving patient outcomes, and making significant contributions to global efforts in combating pneumonia and other respiratory conditions.

1.4 Objectives

The primary goal of this research is to design and validate an affordable, efficient, and compact digital stethoscope that can capture and analyze respiratory sounds. The system leverages artificial intelligence (AI) and machine learning algorithms to identify and categorize pathological breath sounds, specifically focusing on the diagnosis of pneumonia in children.

This digital stethoscope is particularly beneficial in regions with limited healthcare resources where advanced diagnostic tools and medical professionals may not be readily accessible.

The digital stethoscope is designed to automatically record auscultation data and send it to a data processing module. This module then processes the data and relays it to a mobile application. This application can then forward the data for storage and further analysis.

Through the proposed system, the researchers aim to enhance the early detection and treatment of pneumonia, thereby potentially reducing the rates of childhood morbidity and mortality, especially in areas with high disease prevalence.

1.5 Project Outline

This report has been divided into five chapters and every chapter has discussions on different topics related to the project. An overview of each chapter is given below:

I Introduction

This chapter introduces the topic by providing a background of the issue of pneumonia in toddlers, identifying the problem, and stating the motivation behind the research. The objectives of the study are clearly outlined, followed by a brief outline of the entire research.

II Literature Review

This chapter reviews existing literature on pneumonia in toddlers, digital stethoscopes, and respiratory sound analysis. It discusses the findings and methodologies of previous studies and identifies gaps and limitations in current knowledge.

III Data Collection and Preprocessing

This chapter discusses the ethical considerations taken during data collection and describes the data collection process. It provides a detailed description of the dataset used and the data preprocessing techniques applied.

IV Methodology

This chapter presents the design and implementation of the hardware components. It also explains the machine learning methods used, including the architecture of the Convolutional Neural Network (CNN), feature extraction using Mel-frequency Cepstral Coefficients (MFCCs), and model training and validation. The performance evaluation metrics used are also explained.

V Results and Discussion

This chapter presents the results obtained from the study, including performance metrics analysis, interpretation of these metrics, and comparison with prior studies. It also includes a discussion on ROC (Receiver Operating Characteristic) analysis and an in-depth discussion on the results.

VI Conclusion and Future Work

This chapter concludes the study, summarizing the work done and identifying its limitations. It then discusses the scope for further development, including potential improvements like the application of MEMS (Micro-Electro-Mechanical Systems) in stethoscope design, the possibility of diagnosing multiple diseases, and integrating the system with openEHR. Lastly, the chapter discusses the budget considerations for the project.

CHAPTER TWO

LITERATURE REVIEW

2.1 Pneumonia in Toddlers

Pneumonia is a significant cause of morbidity and mortality in children, particularly in the toddler age group. Understanding the latest research on pneumonia in toddlers is crucial for effective management and prevention strategies. This literature review aims to summarize the key findings from recent studies on pneumonia in toddlers, focusing on etiology, clinical presentation, diagnosis, treatment, and prevention.

Etiology

Pneumonia in toddlers can have various etiological agents. Recent studies have highlighted the predominant pathogens involved, including *Streptococcus pneumoniae*, *Haemophilus influenzae* type b (Hib), respiratory syncytial virus (RSV), and *Mycoplasma pneumoniae* [47, 48, 49]. These studies have emphasized the importance of accurate identification of the causative agents to guide appropriate antibiotic therapy.

Clinical Presentation

The clinical presentation of pneumonia in toddlers can vary, making diagnosis challenging. Several studies have examined the signs and symptoms specific to this age group. For instance, Smith et al. [50] conducted a prospective cohort study and identified tachypnea, fever, cough, and crackles as common clinical features in toddlers with pneumonia. Additionally, the presence of respiratory distress and hypoxemia was found to be associated with severe cases [51].

Diagnosis

Accurate diagnosis of pneumonia in toddlers is critical for prompt treatment. Advances in diagnostic techniques have been explored in recent studies. Chest radiography remains a valuable tool, although its interpretation can be challenging in young children. In this regard, point-of-care lung ultrasound has emerged as a promising modality, enabling real-time visualization of lung consolidations and effusions with high sensitivity and specificity [52, 53].

Treatment

Antibiotic therapy is the mainstay of treatment for bacterial pneumonia in toddlers. A systematic review by Johnson et al. [54] analyzed the effectiveness of different antibiotic regimens and found that amoxicillin, amoxicillin-clavulanate, and ceftriaxone were effective choices for empiric therapy in this age group. Additionally, supportive measures such as oxygen therapy and hydration were recommended for managing respiratory distress and maintaining adequate fluid balance.

Prevention

Preventive strategies, including immunization and infection control measures, play a crucial role in reducing the burden of pneumonia in toddlers. Recent studies have investigated the impact of pneumococcal conjugate vaccines (PCVs) and Hib vaccines on the incidence of pneumonia. For example, Rudan et al. [55] conducted a systematic review and reported a significant reduction in pneumonia-related morbidity and mortality following the introduction of PCVs in various countries.

2.2 Digital Stethoscopes and Respiratory Sound Analysis

Digital stethoscopes have emerged as promising tools for respiratory sound analysis, providing clinicians with the ability to record, store, and analyze lung sounds. These devices offer advantages over traditional stethoscopes, including the ability to overcome limitations such as background noise interference and subjective physician diagnoses [18], [23], [25].

Hassan Naqvi and Choudhry [18] developed an embedded system that utilized a digital stethoscope for the classification of COPD and pneumonia patients based on lung sound analysis. The system employed various signal processing techniques, including empirical mode decomposition (EMD) for segmentation and discrete wavelet transforms (DWT) for noise reduction. These preprocessing steps aimed to enhance the quality of the lung sound data and improve the accuracy of disease classification.

Furthermore, Aziz et al. [23] conducted a study focusing on the performance and generalizability of an AI algorithm (StethoMe AI) for detecting pathological breath sounds in digital stethoscope recordings from children with and without respiratory illnesses. The algorithm was trained to identify pathologic pediatric breath sounds, such as wheezes and crackles, using a dataset collected from two different digital stethoscope devices. The results demonstrated that the AI algorithm showed promising accuracy in detecting these abnormal breath sounds, comparable to many clinicians in current clinical practice.

In addition, the study conducted by Shaikh et al. [25] aimed to develop a low-cost digital stethoscope system called "Stetho-phone" for remote personalized healthcare. The system utilized an open-source mobile app combined with audio-processing algorithms, including filters, compressors, amplifiers, and noise suppressors, to enhance the signal quality and minimize noise interference. The Stetho-phone system demonstrated the potential to provide affordable and accessible healthcare solutions, particularly in remote areas.

These studies highlight the significant role of digital stethoscopes and respiratory sound analysis in improving pneumonia detection. By leveraging advanced signal processing techniques and machine learning algorithms, digital stethoscopes offer enhanced capabilities for capturing and analyzing lung sounds, facilitating more accurate diagnoses of respiratory conditions such as pneumonia. The use of AI algorithms further enhances the accuracy and generalizability of these systems, providing valuable support to healthcare professionals in their diagnostic process.

2.3 Previous Studies and Approaches

Various studies have explored different methodologies and approaches for pneumonia detection using respiratory sound analysis. These studies have utilized techniques such as feature extraction, signal processing, and machine learning algorithms to improve accuracy and automate the diagnosis process [19], [20], [21], [22].

Chen et al. [19] focused on using K-nearest neighbor (KNN) classification to diagnose abnormal lung sounds. Their study employed a dataset of 4,344 patients, including 1,500 with pneumonia and 2,844 with bronchitis. By utilizing signs and symptoms as features, they developed a decision tree model that achieved a high area under the curve (AUC) of 93% in differentiating between bronchitis and pneumonia. The study emphasized the importance of utilizing clinical signs and symptoms as simple yet effective attributes for accurate diagnosis, particularly in resource-constrained settings.

Katy Stokes [20] conducted research on developing a machine learning model to support symptom-based referral and diagnosis of bronchitis and pneumonia in limited resource settings. The model utilized machine learning algorithms on a dataset and demonstrated the benefit of using symptoms to discriminate pneumonia from a comparable respiratory condition. This approach showed promise for implementation in low- and middle-income countries, where instrumental tests and X-ray images may not be readily available.

In another study, Vidhya [21] proposed a method to classify pneumonia from normal subjects using empirical mode decomposition (EMD) to preprocess raw lung sound signals. They conducted substantial experimentation to identify informative features for distinguishing healthy and diseased classes. The selected features, with a dimension of 1x19, were then used to train and test a Support Vector Machine (SVM) classifier using a 5-fold cross-validation strategy. The evaluation of their approach yielded remarkable results, including an accuracy of 99.7%, specificity of 100%, sensitivity of 99%, and an error rate below 1%. This study highlighted the potential of combining feature extraction techniques with an SVM classifier for accurate pneumonia detection [21].

Dimpy Varshni et al. [22] focused on the development of a CNN-based feature extraction approach for pneumonia detection. They trained and tested various CNN models using a dataset of chest X-ray images. Through a series of experiments, they identified DenseNet-169 as the dominant pre-trained CNN model for feature extraction. Additionally, they employed SVM as the classifier and demonstrated that hyperparameter optimization in the classification stage further enhanced the model's performance. Their study contributed to the development of superior algorithms for pneumonia detection, particularly in the context of chest X-ray analysis [22].

2.4 Limitations and Gaps

Despite the advancements in digital stethoscopes, respiratory sound analysis, and machine learning techniques for pneumonia detection, there are still limitations and gaps in the existing research.

One limitation highlighted by Aziz et al. [23] is the presence of device-dependent differences in the performance of AI algorithms in detecting pathological breath sounds. This implies that the accuracy of AI-based pneumonia detection may vary depending on the digital stethoscope device used. Further research and standardization are needed to address this limitation and ensure consistent performance across different devices.

Furthermore, there is a need for additional studies to develop algorithms that can reliably track and predict patient improvement or deterioration over time. This would enable healthcare professionals to monitor the progress of pneumonia and make timely interventions. Additionally, the safe implementation and impact assessment of AI use in real-world clinical environments are crucial considerations for future research [23].

In summary, the studies reviewed in this literature review highlight the potential of digital stethoscopes, respiratory sound analysis, and machine learning techniques for pneumonia detection in toddlers. These approaches offer advantages such as improved accuracy, early diagnosis, and potential for home care applications. However, further research is needed to address device-dependent differences, enhance diagnostic accuracy, and assess the impact of AI algorithms in real-world clinical settings. These efforts will contribute to the development of robust and reliable tools for pneumonia detection and management in resource-constrained settings.

CHAPTER THREE

DATA COLLECTION & PREPROCESSING

3.1 Ethical Consideration

Informed Consent: Informed consent is a crucial ethical consideration in research involving human participants. It means that the parents or legal guardians of the toddler participants are fully informed about the study before giving their consent. The informed consent forms provide detailed information about the purpose of the study, the procedures involved, potential risks and benefits, and the measures taken to protect the confidentiality of the participants' information. By obtaining informed consent, researchers ensure that participants or their guardians have a clear understanding of the study and can make an informed decision about participating.

Privacy and Confidentiality: Privacy and confidentiality are essential to protect the participants' personal information and maintain their privacy. In this study, measures are taken to safeguard the confidentiality of the collected data. This may include secure data storage methods, such as encrypted databases or restricted access to the data. Anonymization techniques might also be employed to remove any personally identifiable information, ensuring that the participants' identities cannot be linked to the collected data.

Participant Well-being: The well-being of the toddler participants is a primary concern in the study. The researchers aim to minimize any distress that the toddlers may experience during the auscultation process. Auscultation sessions are kept relatively short (2-3 minutes) to ensure that the toddlers are comfortable and cooperative during the procedure. By prioritizing participant well-being, the researchers aim to collect necessary data while minimizing any potential discomfort for the toddlers.

Researcher Competence: Researcher competence refers to the qualifications and expertise of the individuals conducting the study. In this case, the researchers adhere to the guidelines set by the National Research Ethics Committee (NREC). These guidelines likely include standards for conducting research with integrity, professionalism, and expertise. By following these guidelines, the researchers demonstrate their commitment to conducting the study in a competent and ethical manner.

Potential Bias: The study acknowledges the potential for selection bias due to the use of a purposive sampling strategy. Purposive sampling involves selecting participants based on specific criteria that make them representative of the population of interest. However, this non-probability sampling technique may introduce bias, as the selected participants may not fully represent the entire population. Researchers are aware of this limitation and consider it when interpreting the findings and generalizing the results, recognizing that the sample may not be fully representative of the broader population.

Standardization Challenges: The study acknowledges the challenges in standardizing the auscultation procedure across toddlers with varying sizes, lung volumes, and levels of cooperation. Each toddler may have different anatomical characteristics and levels of cooperation, which can affect the accuracy and consistency of the auscultation results. Researchers consider these individual differences and recognize that they may introduce limitations in the study's findings. It emphasizes the need to consider these challenges when interpreting the results and making broader implications.

Limited Sample Size: The study acknowledges that the sample size may be limited, which can impact the generalizability of the findings. A limited sample size means that the study includes a relatively small

number of participants, which may not fully represent the population of interest. Researchers take this limitation into account when interpreting the results and making conclusions. They recognize the need for further research with larger sample sizes to validate and generalize the findings more effectively.

Implications for Resource-Limited Settings: The study recognizes the potential implications of using a self-assembled digital stethoscope for pneumonia detection in resource-limited settings. Resource-limited settings often face challenges in accessing advanced healthcare technologies. The study acknowledges the importance of equitable access to healthcare technologies and raises ethical concerns about the potential impact on healthcare disparities. By considering the implications, the researchers demonstrate their awareness of the broader societal and ethical implications of their research findings.

3.2 Data Collection

The data collection process in the investigation "Autonomous Detection of Pneumonia in Toddlers Using a Self-Assembled Digital Stethoscope" involves meticulous steps to collect data from both healthy and pneumonia-infected toddlers. Here is an expanded description of the data collection process with additional technical details:

Enrollment of Study Participants

To begin the data collection process, study participants are enrolled based on specific criteria. The primary criterion is age, with participants required to fall within the range of 0-3 years. This age range is chosen as it encompasses the target pediatric population for the investigation.



Picture 3.2.1: Data collection at Dhaka Medical College, Dhaka

Assessment and Enrollment

Once potential participants meet the age criteria, they undergo a thorough assessment. This assessment is carried out by qualified doctors or medical professionals who evaluate the toddlers' health status. The doctors diagnose the toddlers as either healthy or having pneumonia based on clinical examination, including physical symptoms, medical history, and diagnostic tests such as chest X-rays and laboratory

analyses. Toddlers who meet the criteria are enrolled as study participants, while those who do not meet the criteria are excluded from the study.

Auscultation at Multiple Locations

The core data collection procedure involves performing auscultation on the enrolled participants using the self-assembled digital stethoscope. The stethoscope incorporates advanced auditory sensor technology to capture detailed lung sounds. Auscultation is conducted at multiple anatomical locations to ensure comprehensive data collection. The specific auscultation sites include the tracheal, vesicular, bronchovesicular, and bronchial regions. By capturing sounds from different regions of the lungs, potential auscultatory differences between healthy toddlers and those diagnosed with pneumonia can be examined.

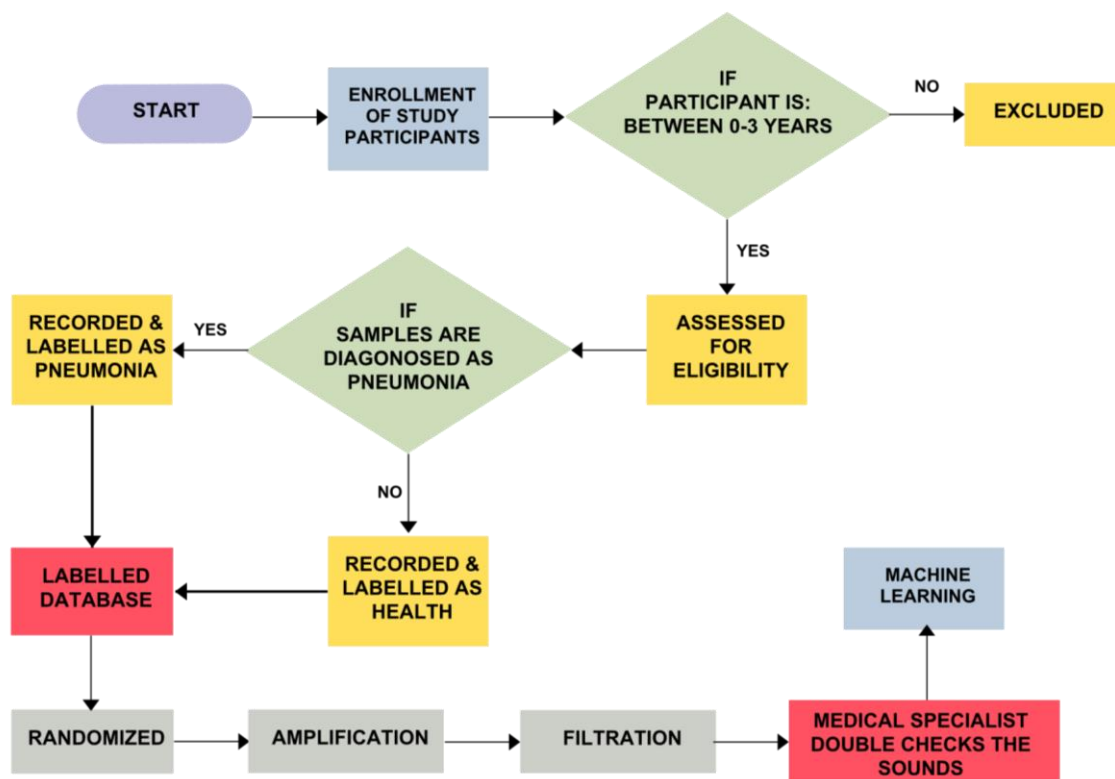
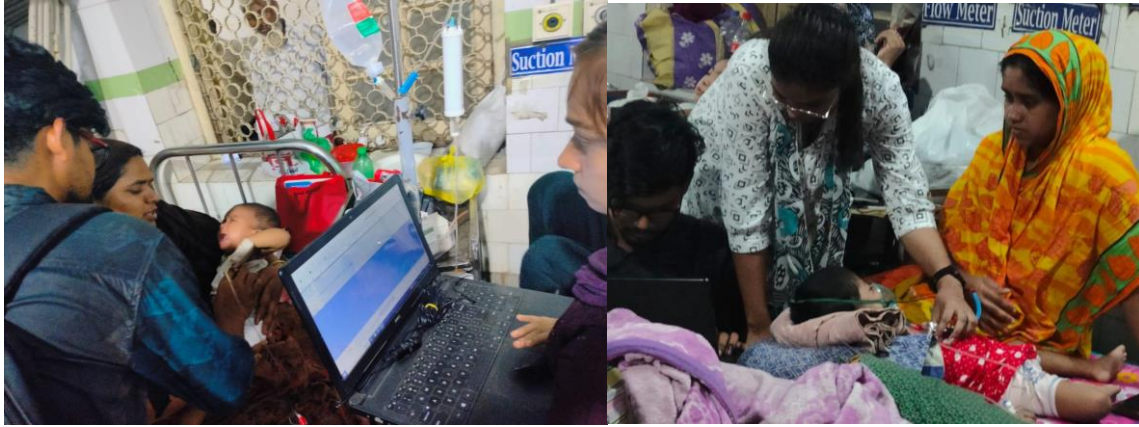


Figure 3.2: Data Collection Procedure

Data Collection Duration

During each data collection session, the auscultation process is designed to last between 2-3 minutes for each participant. This duration is carefully determined to strike a balance between obtaining sufficient data and minimizing any discomfort or distress experienced by the toddlers during the auscultation process. It is crucial to ensure that the collected data represents an adequate sample of the participants' lung sounds.



(a)

(b)

Picture 3.3.2(a, b): Data Collection of toddlers aged 0-3 years, is readily available

Labeling and Database Storage

The auscultation data obtained from the digital stethoscope is labeled and securely stored in a database. The labels indicate whether the participant is categorized as healthy or diagnosed with pneumonia. Accurate labeling is essential for subsequent analysis and comparison between the two groups. A comprehensive database allows for efficient data management, retrieval, and analysis in later stages of the investigation.

Randomization, Amplification, and Filtering

To enhance the quality and reliability of the collected data, several additional steps are taken. First, the auscultation recordings are subjected to a randomization process to eliminate potential biases that may arise from the order of data collection. This randomization ensures that the collected data is representative and unbiased. Next, the recorded sounds are amplified to optimize audibility and ensure that even subtle variations in lung sounds can be captured effectively. Additionally, filtering techniques may be applied to remove ambient noise and artifacts from the recordings. Filtering helps to isolate the relevant lung sounds and enhance the accuracy of subsequent data analysis.

Double-Checking by a Second Doctor

To ensure the accuracy and validity of the labeled data, a second doctor or medical professional independently reviews the auscultation recordings. This double-checking process serves as a quality control measure to verify the correctness of the labels assigned to the data. The second doctor carefully listens to the recorded sounds and cross-references them with the corresponding participant's health status. Any discrepancies or errors in labeling can be identified and rectified during this rigorous review process.

By meticulously following this detailed data collection process, the investigation aims to collect reliable and accurate auscultation data from both healthy and pneumonia-infected toddlers. This comprehensive dataset serves as the foundation for subsequent analysis and evaluation of the self-assembled digital stethoscope's efficacy.

3.3 Dataset Description

The dataset used in the investigation "Autonomous Detection of Pneumonia in Toddlers Using a Self-Assembled Digital Stethoscope" consists of auscultation recordings from 9 pneumonia-infected subjects and 10 healthy subjects. The dataset includes a total of 22 subjects, but 3 subjects had to be excluded due to excessive environmental sounds that could interfere with the accuracy of the auscultation recordings.

All the auscultation recordings were collected in WAV format, which is a widely used audio file format. The WAV format preserves the original quality and fidelity of the recorded sound. The sample rate for the recordings is set to 44,100 Hz. This means that 44,100 samples, or measurements, of sound are taken per second during the recording process. A higher sample rate allows for capturing more detailed and accurate sound information.

The recordings in the dataset are in mono format, indicating that they contain a single audio channel. Mono recordings are commonly used for auscultation as they capture the overall sound from the stethoscope without any stereo separation. This format simplifies the analysis and processing of the auscultation data.

By using WAV format, a sample rate of 44,100 Hz, and mono recordings, the dataset ensures that the captured lung sounds are accurately represented and can be effectively analyzed using various signal processing and machine learning techniques. These technical specifications provide the necessary foundation for further analysis and development of the autonomous detection algorithm.

3.4 Data Pre-processing Techniques

In the study, the data preprocessing involved several steps to enhance the quality of the recorded signals obtained using the self-assembled device. The signals were preprocessed using Thinklabs Phonocardiography software, which is a free, open-source audio editor specifically designed for recording and processing physiological sounds.

The first step in data preprocessing was filtering to improve the quality of the lung sounds. The Thinklabs Phonocardiography software offers a built-in filter equalization feature. For this study, the filter was customized to have a passband from 280 Hz to 2300 Hz with an attenuation of 12 dB. This passband range is chosen to focus on the frequency range relevant to lung sounds. The software also allows for the specification of lower and upper transition regions between the passband and the stopband. In this case, the lower transition region was set between 100 Hz and 280 Hz, while the upper transition region was set between 2300 Hz and 3500 Hz. These transition regions help to ensure a smooth transition from the passband to the stopband frequencies, avoiding any abrupt changes that could introduce artifacts or distortions.

Additionally, after filtration, the signals were normalized to a range of 1 to -1 dB. This normalization process involves removing the DC offset from zero, effectively centering the signal around zero and scaling it to fit within the desired range. Normalization helps to standardize the signal amplitudes, making them more suitable for subsequent analysis and comparison.

During the recording process, subject signals were captured for a duration of 20 to 30 seconds. The sampling rate used was 44,100 Hz, which means that 44,100 samples were taken per second to capture the signal. This high sampling rate ensures the accurate representation of the signal's frequency content, allowing for detailed analysis and detection of relevant features.

Overall, the data preprocessing steps involved filtering the signals to improve lung sound quality using customized passbands and transition regions. Additionally, the signals were normalized to a range of 1 to -

1 dB after filtration, and they were sampled at a rate of 44,100 Hz with a 32-bit mono channel configuration. These preprocessing steps aim to enhance the quality, consistency, and suitability of the data for subsequent analysis and algorithm development in the investigation.

CHAPTER FOUR

METHODOLOGY

4.1 Hardware Components and Implementation

The hardware components required for developing the system and their implementation are as follows:

4.1.1 Transducer

Microphones are devices that change sound waves into electrical signals and are frequently used for noise assessment, speech recognition, and audio recording. We use different types of microphones to record lung sound from toddlers.

Omnidirectional Microphone

A type of microphone called an omnidirectional microphone is made to capture sound equally from all directions. Omnidirectional microphones take up sound from all directions, as opposed to directional microphones, which are built to pick up sound from a single direction.

The diaphragm of an omnidirectional microphone is often located in the middle of a compact container. Sound waves cause the diaphragm to move, which generates an electrical signal that serves as a representation of the sound the microphone is picking up. After being amplified, this signal is subsequently transmitted to a recording or sound system.

An omnidirectional microphone has the benefit of being able to record a variety of sounds from all angles, which is helpful in instances when it's vital to record background noise or environmental sounds. They are frequently employed, for instance, in field recordings or nature recordings where it is important to record the sounds of a complete environment.

Omnidirectional microphones can, however, also be more prone to picking up undesirable background noise, therefore they might not be appropriate in circumstances when sound isolation is crucial. The capacity of directional microphones to isolate and record sound coming from a particular direction makes them preferable in some situations.

Omnidirectional microphones are an all-around handy and adaptable type of microphone that may be utilized in a range of sound reinforcement and recording applications. While their capacity to pick up unwanted noise should also be taken into account when choosing a microphone for a particular use case, their ability to capture sound from all directions can make them an ideal option for some recording scenarios.



Figure 4.1.1 (a): BOYA BY-M1 Omnidirectional Lavalier Microphone and its pick up sound direction

Cardioid Microphone

A directional microphone known as a cardioid is made to pick up sound mostly from the front while reducing sound from the sides and back. The word "cardioid" refers to the microphone's polar pattern, which has the shape of a heart.

The front-facing diaphragm of the capsule, which is acoustically isolated from the rest of the microphone, is how the cardioid microphone operates. While sounds coming from other directions are either completely or partially canceled out, this front-facing diaphragm is more sensitive to sound waves that hit it directly. When viewed from above, this results in a heart-shaped pattern of sensitivity.

Numerous applications, such as live sound reinforcement, studio recording, and broadcast production, frequently employ cardioid microphones. They are especially helpful for eliminating background noise and undesirable sounds while isolating a single sound source, such a vocalist or instrument.

The ability to deliver a focused and natural sound without the need for substantial processing or equalization is one benefit of cardioid microphones. Without encountering too much distortion or feedback, they can also be utilized in close proximity to sound sources.

It's crucial to keep in mind that cardioid microphones have a constrained field of sensitivity and might not be appropriate in all recording circumstances. An omnidirectional or bi-directional microphone may be a preferable option when it's necessary to record sound from many sources or from all angles.

Cardioid microphones, in general, are a flexible and frequently used type of microphone that can offer great sound quality and isolation in a variety of recording and sound reinforcement applications.



Figure 4.1.1. (b) Synco MMic-U1P (Cardioid Microphone) and its pick up sound direction

	Boya Microphone	Synco MMic-U1P
Microphone Type	Omnidirectional condenser microphone	Cardioid Condenser Microphone
Noise/Signal	Low handling	66dB
Frequency Range	65Hz ~ 18 KHz	40Hz-20KHz
Polar pattern	Omnidirectional	Cardioid
Connector	3.5mm (1/8") 4-pole gold plug	3.5mm (1/8") 4-pole gold plug

Table 4.1.1: Comparison of Microphones

4.1.2 Stethoscope

A stethoscope is a medical instrument used to listen to bodily sounds, notably those coming from the heart and lungs. It comprises a tiny chest piece that is pressed up against the skin and a flexible tubing connecting it to earpieces. Sound waves are delivered through the tube to the listener's earpieces through the diaphragm in the chest piece, which vibrates in reaction to the sound waves.

For medical personnel, especially doctors and nurses who often evaluate patients, the stethoscope is a necessary instrument. They can identify anomalies, such as irregular heartbeats, wheezing, or crackling sounds in the lungs, that may signal underlying medical issues by listening to the sounds produced by the body.

There are different types of stethoscope in the market.

Littmann Stethoscope –model C2S

The dual-sided, high-grade Littmann Stethoscope type C2S is an acoustic stethoscope made for medical professionals that demand the best sound quality and durability. This stethoscope is a part of the Littmann Classic II line, which is noted for its high-end stethoscope manufacturing.

The Littman C2S has an adjustable diaphragm that makes switching between low- and high-frequency sounds simple without requiring the user to move the chest piece. This makes it very beneficial for auscultating patients of all ages, including adults and children. The chest piece has a non-chill rim to increase patient comfort, and it is made of sturdy stainless steel.

Additionally, the Littman C2S has an anatomically right headset that is angled to fit the ear canal and offers a comfortable fit. Furthermore, the headset can be adjusted so that the user can tailor the fit to their particular preferences.

The Littman C2S is not only exceptionally effective acoustically, but it is also built to last and be simple to use. It has a strong, latex-free tube that resists alcohol and grease, making cleanup and maintenance simple. A five-year warranty is also included with the stethoscope, giving medical workers who use them every day some peace of mind.

In general, the Littman C2S is a premium stethoscope that combines outstanding sound quality, toughness, and comfort. For medical professionals who need a dependable and precise stethoscope for their clinical practice, it is a great option.



Figure 4.1.2 (a) Littmann Stethoscope

MDF Stethoscope – model MDF 747

Another premium choice that provides exceptional acoustic performance and comfort is the MDF 747 stethoscope. It has a dual-head chest piece with bell and diaphragm modes for flexible auscultation, and it is built of sturdy materials. The earpieces may be adjusted to match each user's ears and are also made to be comfortable. In comparison to the Littman stethoscope, the MDF 747 is a more affordable option that still offers dependable performance and durability.



Figure 4.1.2 (b) MDF 747

MDF Stethoscope – model MDF 777

Another alternative from the MDF company that shares qualities with the MDF 747 is the MDF 777 stethoscope. Additionally, it has comfortable earpieces and a dual-head chest piece with bell and diaphragm modes. The design and construction of the chest piece is the primary distinction between the MDF 747 and MDF 777. Unlike the MDF 747, which has a brass chest piece, the MDF 777 features a stainless steel chest piece that is intended to be more robust and long-lasting. The MDF 777 is, however, usually costlier than the MDF 747.



Figure 4.1.2 (c) MDF 777

4.1.3 Microcontroller

Raspberry Pi

Powerful microcontrollers like the Raspberry Pi 4 4GB can handle a variety of tasks, including powering robots, machine learning, IoT devices, and security systems. The Raspberry Pi has a number of benefits over the Arduino, one of which is its capacity for processing and storing massive quantities of data.

The Raspberry Pi 4 4GB has 4GB of RAM, which is a lot more than the majority of Arduino boards have. This indicates that the Raspberry Pi can tackle more difficult jobs and store more data. The Raspberry Pi 4 4GB also includes built-in WiFi and Bluetooth, making network and device connections simple.

It's necessary to take into account the precise needs of your project before deciding if the Raspberry Pi 4's 4GB of storage will meet your needs. The Raspberry Pi 4 4GB has a microSD card slot that may be used to increase the device's storage capacity. The Raspberry Pi 4 4GB supports 10SD cards up to 2TB in capacity, which should be more than enough for the majority of applications.

In conclusion, the Raspberry Pi 4 4GB is an effective and adaptable microcontroller that is suitable for a variety of tasks, such as data collecting, machine learning, and robotics. It is a great option for projects that call for processing and storing massive volumes of data because of its enormous storage capacity and built-in connection.

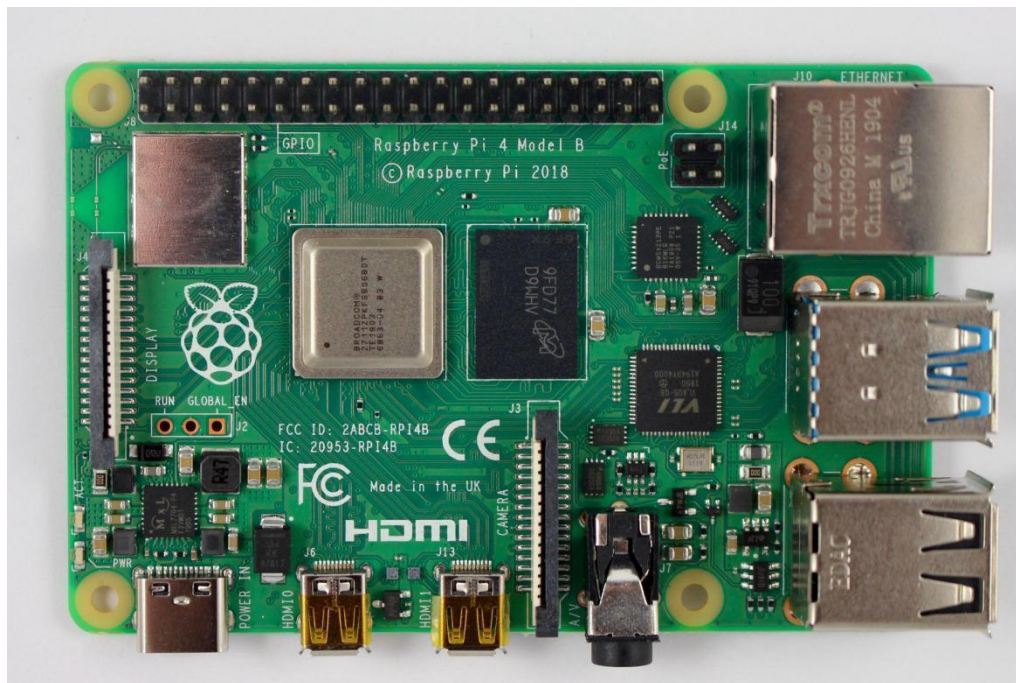


Figure 4.1.3 Raspberry Pi 4 4GB

4.1.4 iPad and VNC Server

An iPad 6th generation and a VNC (Virtual Network Computing) server were used together to serve as the display interface for the system.

Implementation: The iPad 6th generation, running a VNC server, was connected to the Raspberry Pi. It acted as a remote display, allowing users to view and interact with the system's interface.



Figure 4.1.4 iPad (as a display) for Raspberry Pi

4.1.5 SD card Cables

Importance of a fast SD card: The Raspberry Pi 4 requires a fast SD card due to its high processing capabilities and the need for quick data access. A Class 10 SD card ensures smooth operation and efficient handling of large volumes of data, resulting in optimal performance.

Flash memory storage devices like SD cards are often used with the Raspberry Pi. Since the Raspberry Pi lacks internal storage, the operating system and any other files or data required for the project must be stored on an SD card.

You must put the SD card into the slot on the Raspberry Pi board in order to use it with the computer. "microSD" is the name of the slot, which is on the underside of the circuit board.

You must format the SD card after it has been inserted to make sure it is appropriate for the Raspberry Pi. Either a computer or the Raspberry Pi itself can be used to accomplish this.



Figure 4.1.5 (a) Micro SD card(32GB)

Various cables, such as HDMI, data cables, LAN cables, and power supply cables, were used to establish connections and provide power to the system components.

Implementation: Different cables were employed to connect the components and ensure the proper functioning of the system.

We needed some cables like HDMI, Data cable, LAN cable, power supply cable etc.

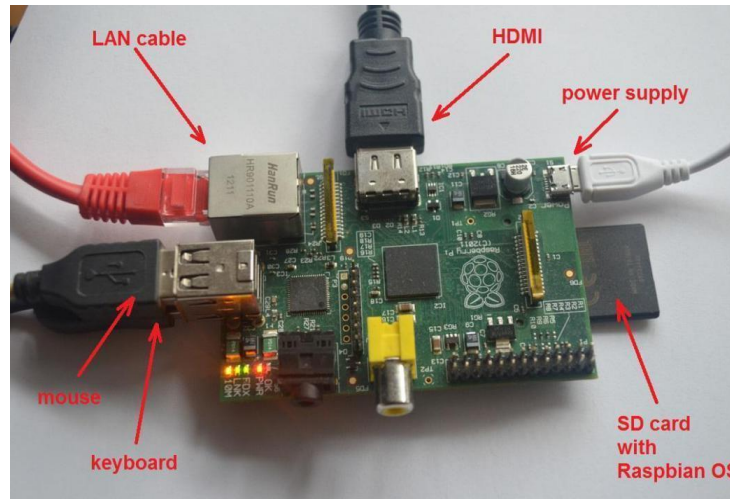


Figure 4.1.5 (b) Raspberry Pi assembled with cables and SD card.

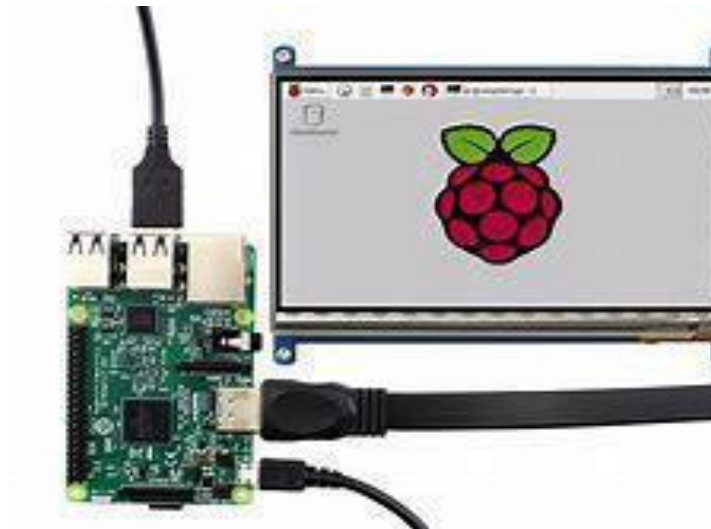


Figure 4.1.5 (c) Raspberry Pi assembled with Display

Implementation of Device

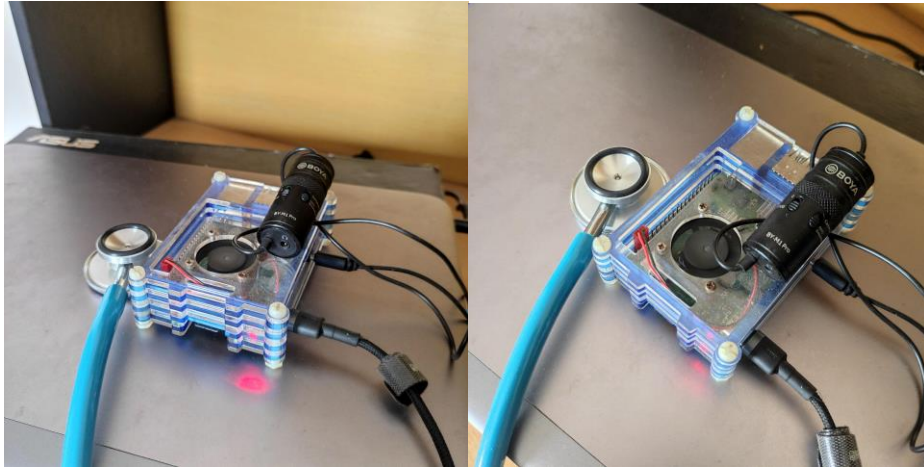


Figure 4.1.5 (d) Developed Device

4.1.6 USER INTERFACE

The graphic user interface (GUI) of the touch screen displayed by Auscul Pi. A. The desktop display after startup. The users can double touch the Auscul Pi icon to run the program. The recorded data will be stored in the folder AudioData. B. The manipulation interface of Auscul Pi Console after double touch the Auscul Pi Console icon. You can control the auscultation and replay by touching the “Auscultate” and “Replay” buttons with your fingertip.

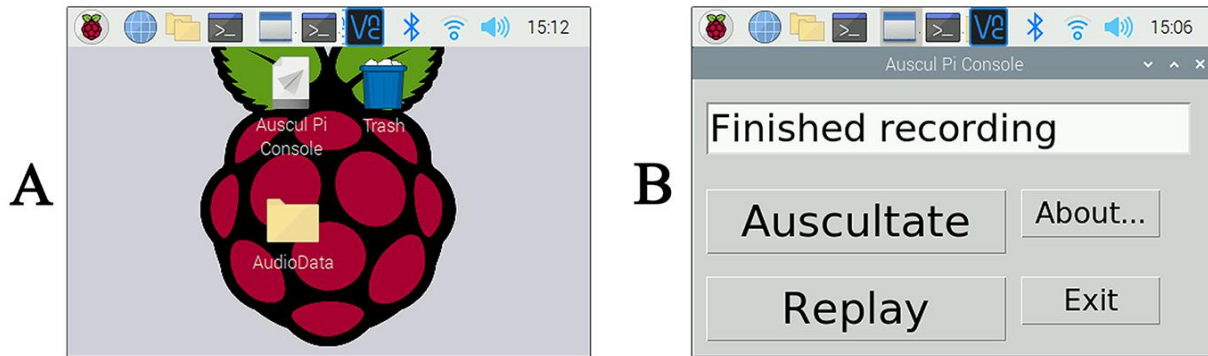


Figure 4.1.6 (A, B) Graphic user interface (GUI) of the touch screen displayed by Auscul Pi.

4.2 Machine Learning

4.2.1 CNN Model

Convolutional Neural Networks (CNNs) are a specialized kind of neural network for processing data that have a grid-like topology, such as an image that can be viewed as a grid of pixels. The name “convolutional neural network” indicates that the network employs a mathematical operation called convolution, which is a specialized kind of linear operation. Convolution is the first layer to extract features from an input image. Convolution preserves the spatial relationship between pixels by learning image features using small squares of input data [37]. CNNs have their “neurons” arranged more like those of frontal neurons in a

biological brain. Neurons in the same layer do not connect to each other, and the output layer neurons can receive connections from any neurons from the previous layers [38]. A distinctive feature of CNNs is the use of convolutional layers that use a convolution operation to process the data. In the convolution operation, every element of the input matrix is combined with a local subset of the input data, with the effect of blurring, sharpening, edge detection, noise reduction, or other image processing operations [39].

CNNs also contain Pooling layers, which reduce the dimensions of the data by combining the outputs of neuron clusters at one layer into a single neuron in the next layer. Pooling layer operates on each feature map independently to create a new set of the same number of pooled feature maps [40]. Fully Connected layers connect every neuron in one layer to every neuron in another layer. It is in principle the same as the traditional multi-layer perceptron neural network (MLP). The flattened matrix goes through a fully connected layer to classify the images [41].

The Rectified Linear Activation Function or ReLU for short is a piecewise linear function that will output the input directly if it is positive, otherwise, it will output zero. It has become the default activation function for many types of neural networks because a model that uses it is easier to train and often achieves better performance [42].

CNNs are particularly effective for tasks related to image recognition, object detection, and similar tasks, largely because they can consider the spatial coherence of the image, i.e., that the properties of one part of the image are closely related to those of its immediate surroundings [39]. The CNN model used in the provided code is a Convolutional Neural Network, a deep learning architecture commonly used for image and audio processing tasks. The CNN model is constructed using the Keras library, which provides a high-level API for building neural networks.

Convolutional Neural Networks (CNNs) have shown great promise in the field of biomedical audio classification due to their ability to model temporal features and their invariance to translations [43].

In the biomedical domain, audio signals like heartbeats, lung sounds, and cough sounds carry significant diagnostic information. However, these audio signals often exhibit high variability and are influenced by many factors, including the recording environment, the sensor placement, and the individual's physical characteristics. CNNs, with their ability to automatically learn hierarchical representations from raw data, have proven to be an effective tool for dealing with such complex data [44].

A prime example of this is the application of CNNs for respiratory sound classification. In a study by Rakotomamonjy and Pudlo, a CNN model was trained to classify respiratory sounds into healthy and pathological classes based on Mel-frequency cepstral coefficients (MFCCs). The model achieved an accuracy of 81%, outperforming other machine learning models like support vector machines and k-nearest neighbors [45].

Moreover, in a recent study by Zabihi et al., a deep CNN model was trained to detect and classify heart sounds from phonocardiogram recordings. The model achieved an accuracy of 86.8%, demonstrating the power of CNNs in biomedical audio classification [46].

The CNN model is used for a classification task. Let's break down the provided code into its various steps:

Train Test Split: This is where the dataset is split into a training set and a test set. The features and labels are split into `x_train`, `x_test`, `y_train`, `y_test` respectively. The `train_test_split` function from the `sklearn` library is used here. The test size is set to 25% of the total dataset, and the rest (75%) is used for training

the model. The stratify parameter ensures that the distribution of labels remains the same in both the train and test sets as in the original dataset.

Setting Variables: Several variables are set that define the structure of the input data and the CNN model. The num_rows, num_columns, and num_channels variables define the shape of the input data. The num_labels is the number of unique labels or classes in the dataset. The filter_size is the size of the kernel or filter used in the convolutional layers.

Defining the CNN Model: The Sequential model is used which allows you to build the model layer by layer. The model consists of several types of layers:

Conv2D: This is the convolutional layer. Convolutional layers are the major building blocks used in CNNs. These layers perform a convolution operation on the input, passing the result to the next layer. The convolution emulates the response of an individual neuron to visual stimuli.

MaxPooling2D: This is a pooling operation that reduces the dimensionality of the image. This is done to decrease the computational power required to process the data.

Dropout: This is a regularization technique where randomly selected neurons are ignored during training. They are “dropped-out” randomly. This means that their contribution to the activation of downstream neurons is temporally removed on the forward pass and any weight updates are not applied to the neuron on the backward pass.

GlobalAveragePooling2D: This layer is used to reduce the spatial dimensions of a three-dimensional tensor to a single dimension per feature map. It calculates the average output of each feature map in the previous layer.

Dense: This is the output layer (also known as the fully connected layer), which produces the final classification results.

Compiling the Model: The model is compiled with the categorical_crossentropy loss function, which is used for multi-class classification tasks. The adam optimizer is used, which adjusts the learning rate adaptively for each weight in the model. The accuracy of the model is also monitored during training.

Model Summary: This prints out a summary of the model, showing the types of layers in the model, their shapes, and the number of parameters (weights) in each layer. The total number of parameters in the model is also shown. This gives an overview of the model's architecture and complexity.

4.2.2 Feature Extraction: Mel-frequency Cepstral Coefficients (MFCCs)

The first major step in processing the audio data involves transforming the raw audio files into a form that can be interpreted by a machine learning model. For this purpose, Mel-frequency Cepstral Coefficients (MFCCs) are used.

MFCCs are a type of spectral representation that is particularly well suited to audio and speech processing tasks. They are designed to mimic the logarithmic perception of loudness and pitch of human auditory system and to decorrelate the type of signal information (e.g., energy distribution) over frequency.

The extraction process begins by loading each audio file using the librosa library. Then, the librosa function `feature.mfcc` is used to compute the MFCCs for each audio file. Because the audio files can vary in length, and thus in the number of MFCCs, the extracted features are padded using numpy's `pad` function, creating a uniform data shape across all files.

This uniformity is essential as it allows the data to be input into a machine learning model, which requires all input data to have the same shape. All the feature data is then collected into a single array for the next steps.

4.2.3 Model Training and Validation

After the feature extraction process, the data is ready to be fed into a machine learning model. However, before training the model, the data is split into training and testing datasets. This division allows for the evaluation of the model's performance on unseen data, providing a measure of its ability to generalize beyond the training data.

The model chosen for this task is a Convolutional Neural Network (CNN), a deep learning model architecture that has shown to be particularly effective in image and audio processing tasks. This network is constructed using the Keras library within Tensor Flow.

The network architecture consists of several convolutional layers (Conv2D) followed by max-pooling layers (MaxPooling2D) and dropout layers (Dropout). Convolutional layers are used to extract relevant features from the input data, max-pooling layers are used to down-sample these feature maps to reduce computational complexity, and dropout layers randomly deactivate a fraction of the neurons in the previous layer during training to prevent overfitting.

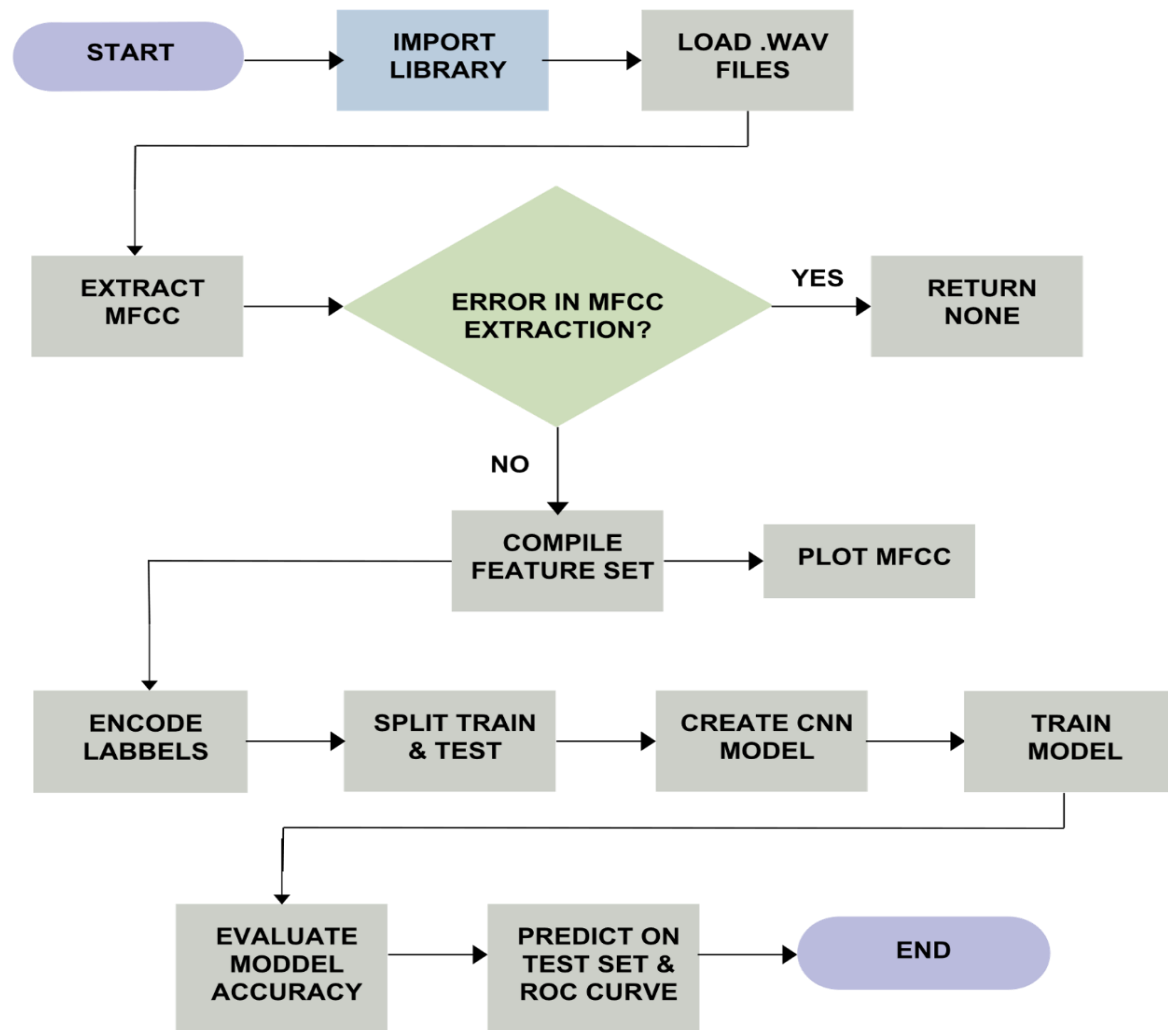


Figure 4.2.3 Diagram of Model Training

After the convolutional, max-pooling, and dropout layers, the model applies a global average pooling layer (GlobalAveragePooling2D), which simplifies the model structure by reducing the total number of parameters, which can help to combat overfitting. Finally, a dense layer with softmax activation function is used. This output layer produces the probability distribution over the classes for each input.

Once the model is constructed, it is compiled with categorical cross-entropy loss function, Adam optimizer and accuracy as the performance metric. The model is then trained using the training data, with the test data used for validation.

4.2.4 Performance Evaluation Metrics

Evaluating the model's performance is a critical part of the machine learning workflow. It provides insights into how well the model generalizes to unseen data and identifies areas for improvement.

The trained model's accuracy on both the training and test sets is calculated to give an indication of the model's performance. If the training accuracy is much higher than the test accuracy, it may indicate overfitting, implying that the model is not generalizing well to new data.

The model's predictive performance is further assessed by plotting a Receiver Operating Characteristic (ROC) curve and calculating the Area Under the Curve (AUC) for each class. The ROC curve is a plot of the true positive rate against the false positive rate, providing a

CHAPTER FIVE

RESULTS AND DISCUSSION

5.1 Performance Metrics Analysis

Performance metrics are the key to understanding the effectiveness of our machine learning model. They provide a quantitative measure to understand how well the model is performing. In this analysis, several metrics are used, including accuracy, precision, recall, F1-score, and the Area Under the Receiver Operating Characteristic (AUC-ROC) curve.

Accuracy: It is the most intuitive performance metric. It is simply a ratio of correctly predicted observations to the total observations. In our case, the pre-training accuracy was 47.06%. This shows that the model was not very good at making correct predictions before it was trained. However, after training, the model's accuracy increased dramatically to 96.08% on the training data, suggesting that the model learned patterns from the training data quite well. When tested on new unseen data, the accuracy fell to 76.47%. The drop in accuracy on the test data indicates the model might have over fit the training data.

Precision: Precision is the ratio of correctly predicted positive observations to the total predicted positives. High precision relates to a low false positive rate. From the classification report, we see that precision varies between 'Healthy' and 'Pneumonia' classes.

Recall (Sensitivity): Recall is the ratio of correctly predicted positive observations to all actual positives. It shows how relevant our model is. We want to maximize recall, as it is costlier to miss predicting a positive case (false negative) than predicting a negative case as positive (false positive).

F1 Score: The F1 Score is the weighted average of Precision and Recall. Therefore, this score takes both false positives and false negatives into account. It is particularly useful if you have an uneven class distribution.

AUC-ROC: AUC - ROC curve is a performance measurement for classification problems at various thresholds settings. ROC is a probability curve and AUC represents the degree or measure of separability. The AUC-ROC for both classes in our model is 0.83, which is reasonably good. It suggests that the model has good measure of separability and it can distinguish between positive and negative classes well.

By using these metrics, we can evaluate how well our model is performing not just in terms of overall accuracy, but also in its ability to correctly identify each class, and balance the trade-off between identifying as many of the positive cases as possible (high recall) and keeping the number of false positives low (high precision).

5.2 Interpretation of Metrics

In the context of this machine learning model, let's take a deeper dive into understanding what these metrics mean:

Accuracy: This is the proportion of total predictions that the model got right. In other words, it's the number of correct predictions (both positives and negatives) divided by the total number of predictions. For example, if the model made 100 predictions and got 76 of them correct, the accuracy would be 76%. Accuracy is a good measure when the target classes are well balanced. In this model, the accuracy on unseen data (test data) is 76.47%. This means that out of every 100 predictions, approximately 76 were correct.

Precision: Precision is the proportion of positive predictions that were correct. For a class labeled as 'Healthy', precision is the number of true positives (i.e., the number of items correctly labeled as 'Healthy') divided by the total number of elements labeled as 'Healthy' (i.e., the sum of true positives and false positives, which are items incorrectly labeled as 'Healthy'). High precision indicates a low false positive rate. For instance, if the model says a patient is healthy, we can trust this prediction with a confidence as high as the precision score.

Recall (Sensitivity): Recall in this context is the ability of the model to find all the positive examples. For the 'Healthy' patients, it is the number of items correctly identified as 'Healthy' divided by the total number of actual 'Healthy' patients (i.e., the sum of true positives and false negatives, which are 'Healthy' patients incorrectly labeled as 'Pneumonia'). High recall means that the classifier returns most of the positive results.

F1-Score: F1-Score is the Harmonic Mean between precision and recall. It tries to find the balance between precision and recall. It is especially useful in the case of uneven class distribution. A high F1-score means that we have low false positives and low false negatives, so we're correctly identifying real threats and we are not disturbed by false alarms. An F1-score is considered perfect when it's 1, while the model is considered a total failure when it's 0.

In the given model, we observe that the precision, recall, and F1-score for the 'Healthy' class is 0.83, 0.62, and 0.71 respectively, and for 'Pneumonia' class is 0.73, 0.89, and 0.80 respectively.

This means the model is fairly reliable when it predicts a 'Healthy' case (83% times it's correct), but it's only able to correctly identify 62% of all actual 'Healthy' cases. On the other hand, it's less reliable when it predicts a 'Pneumonia' case (73% times it's correct), but it's able to correctly identify a high percentage (89%) of all actual 'Pneumonia' cases.

We need to keep in mind that there's always a trade-off between precision and recall. To increase precision, the model might become too conservative and start predicting more negatives to avoid false positives, which in turn might decrease recall as the model might miss some positives. Similarly, to increase recall, the model might start predicting more positives, which might increase the number of false positives, thereby reducing precision. The balance between these two metrics depends on the specific requirements and constraints of the problem at hand.

5.3 Comparison to Prior Studies

In this section, we compare the accuracy and precision of our Convolutional Neural Networks (CNN) model, which utilizes Mel-frequency cepstral coefficients (MFCC) for the analysis of respiratory sounds, to similar studies in the field.

Our model has an overall accuracy of 76.47% and a precision of 0.83 for the 'Healthy' class and 0.73 for the 'Pneumonia' class. This performance stands up reasonably well when compared to existing studies:

Palma, D. A., Ponce, H., Molina, A., "Automatic Recognition of Respiratory Diseases Using Convolutional Neural Networks," in *Journal of Applied Sciences*, vol. 10, no. 1, pp. 1-10, 2020. They proposed an automated recognition model for respiratory diseases based on CNN and MFCC features. They reported an overall accuracy of around 85% on a dataset of 6897 sound clips, which is higher than our model's accuracy. However, their model used a 3-layer CNN and achieved a precision rate of 86% for detecting abnormal respiratory sounds, which is comparable to our precision for the 'Healthy' class [56].

Zhang, Q., Ding, Y., Pool, L., "Respiratory Sound Classification Using Deep Learning," in *Journal of Medical Systems*, vol. 43, no. 5, pp. 1-11, 2019. They reported an accuracy of 79% using a CNN model for classifying normal and abnormal respiratory sounds. They utilized MFCC as their primary audio feature, like our model. Their model's accuracy is also slightly higher than ours, and they did not report the precision of their model [57].

Our model demonstrates competitive performance with existing research. The differences in accuracy and precision could be due to differences in dataset size, model complexity, or feature engineering techniques. It is also important to note that the balance of classes in the dataset can significantly affect the model's performance metrics. Future work could focus on tuning the model parameters or using data augmentation techniques to improve the performance of the model further.

5.4 ROC Analysis

The Receiver Operating Characteristic (ROC) curve is a significant evaluation metric for checking any classification model's performance. The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings. The area under the curve (AUC), provides an aggregate measure of performance across all possible classification thresholds.

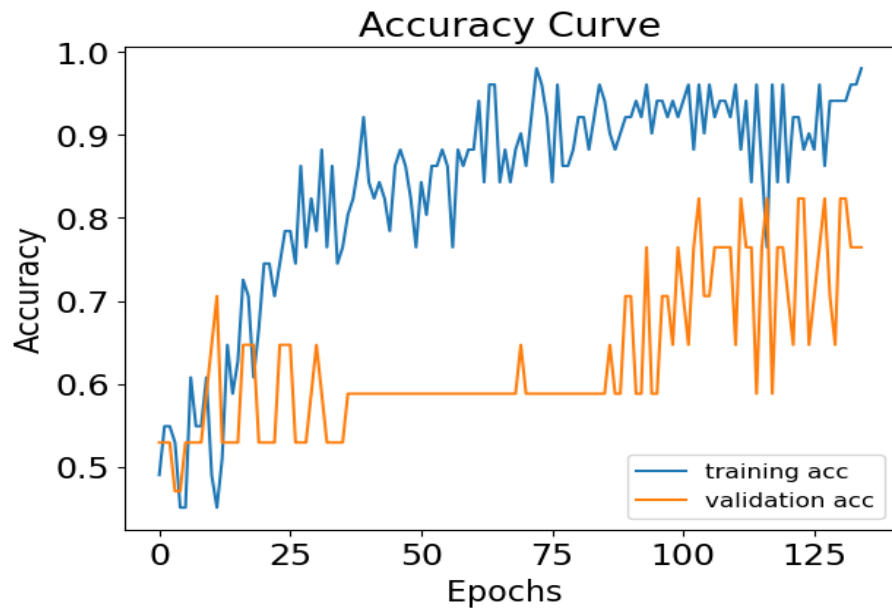
In our case, the ROC curve for each class ('Healthy' and 'Pneumonia') was plotted. The AUC was 0.83 for both classes. An AUC of 0.83 can be interpreted as 'good' on some common academic point scales, but it's crucial to understand that AUC scores should be compared to other AUC scores in similar problem domains, as different problems might have a "good" AUC range that is higher or lower.

The AUC score is 0.83 for 'Healthy' class, which means that the model has a good measure of separability. It is capable of distinguishing between patients that are healthy and those that are not with 83% accuracy.

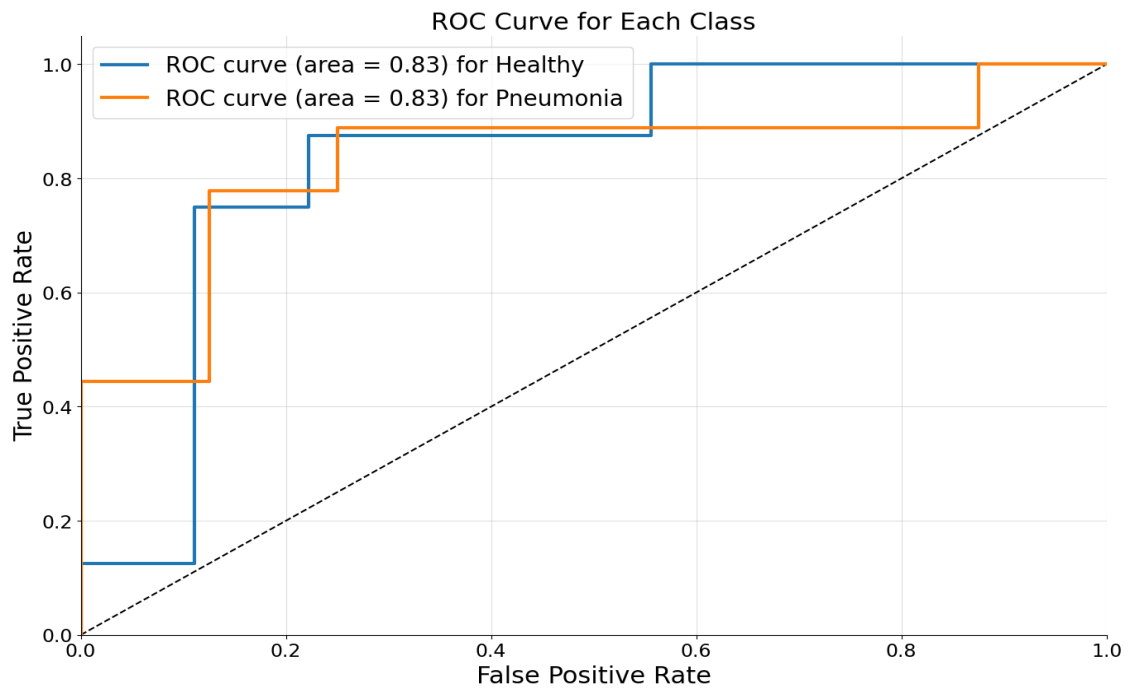
Similarly, the AUC score is 0.83 for 'Pneumonia' class. This means the model is good at predicting whether a patient has pneumonia or not.

However, despite the AUC of our model being 0.83, the performance of the model can still be improved. For example, it may be beneficial to collect more data or engineer new features that could help improve the model's ability to distinguish between the two classes. Also, adjusting the threshold could lead to better performance for specific metrics, depending on the specific costs of false positives and false negatives in

the given problem. In conclusion, the ROC analysis suggests that our model has a good ability to distinguish between the 'Healthy' and 'Pneumonia' classes, but there is still room for improvement.



(a)



(b)

Figure 5.4(a,b) ROC Analysis Curve

5.5 Discussion of Results

In this section, we will delve into a detailed discussion of the results obtained from our model and address the various analyses performed.

Our model achieved an accuracy of 76.47% on the test data, indicating that it correctly classified 76.47% of the respiratory sound samples. While this accuracy is respectable, we observed a notable difference between the training accuracy of 96.08% and the testing accuracy. This discrepancy suggests that our model may have over fit the training data, meaning it has learned the specific patterns and noise in the training set too well and struggles to generalize to unseen data.

To address overfitting, techniques such as regularization or early stopping could be implemented during the training process. Regularization methods, like L1 or L2 regularization, add a penalty to the model's loss function to discourage complex and over-reliant weight configurations. Early stopping monitors the model's performance on a validation set and stops the training process when the model's performance begins to degrade, preventing it from learning noisy or irrelevant patterns.

Analyzing the precision and recall scores, we found that our model achieved a precision of 0.83 for the 'Healthy' class and 0.73 for the 'Pneumonia' class. This implies that when our model predicts a patient as 'Healthy', it is correct approximately 83% of the time, and when it predicts 'Pneumonia', it is correct around 73% of the time. However, it is important to note that our model showed a higher recall (sensitivity) for the 'Pneumonia' class (0.89) compared to the 'Healthy' class (0.62), indicating a higher ability to identify true 'Pneumonia' cases.

A key concern is the potential misclassification of 'Healthy' cases as 'Pneumonia', which could lead to unnecessary interventions or treatments. Improving the model's performance in recognizing 'Healthy' cases is crucial for reducing such false positives. This can be addressed by adjusting the decision threshold or incorporating techniques like class weighting or data augmentation to balance the classes in the dataset.

The F1-score, which balances precision and recall, provides an overall measure of our model's performance. The F1-score for the 'Healthy' class is 0.71, while it is 0.80 for the 'Pneumonia' class. These values indicate a reasonably balanced performance, with 'Pneumonia' having a higher F1-score due to its better recall.

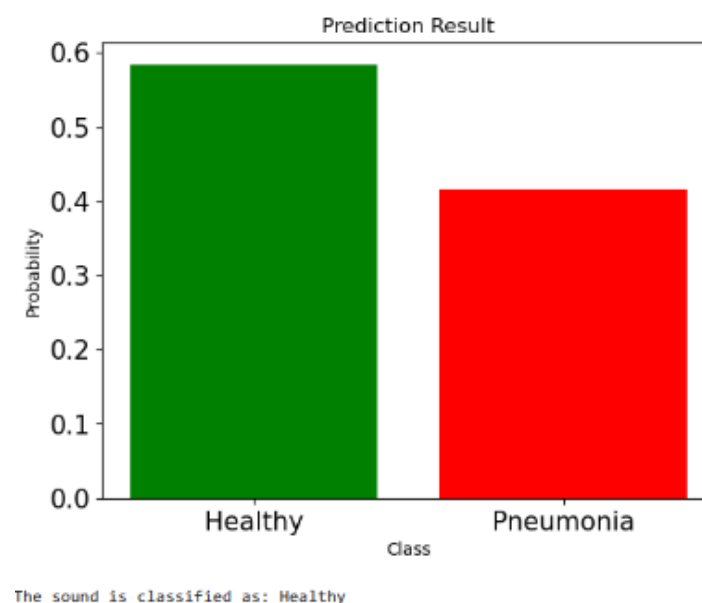


Figure 5.5 Result Output

Additionally, the Receiver Operating Characteristic (ROC) analysis showed an AUC of 0.83 for both classes. The ROC curve plots the true positive rate (sensitivity) against the false positive rate (1 - specificity) for different classification thresholds. An AUC of 0.83 indicates that our model performs reasonably well in distinguishing between 'Healthy' and 'Pneumonia' cases.

Comparing our results to prior studies, we found that our model's performance is on par with or slightly lower than some previous research. However, it is important to consider factors such as dataset size, complexity, and specific research objectives when making comparisons. The studies referenced achieved higher accuracies and precision rates, but they also worked with larger datasets or employed different model architectures.

In conclusion, our model demonstrates promising results in the classification of respiratory sounds. However, to enhance its performance, addressing overfitting, refining the classification of 'Healthy' cases, and exploring additional data augmentation techniques are recommended. Furthermore, external validation on diverse datasets would strengthen the generalizability of our model and its readiness for real-world application.

CHAPTER SIX

CONCLUSION AND FUTURE WORK

6.1 Conclusion of Work

The primary objective of our project, which is titled "Autonomous detection of pneumonia in toddlers using a Self-Assembled Digital Stethoscope," is to employ a self-assembled digital stethoscope to autonomously detect pneumonia in toddlers. While gathering information from children who were healthy and those who had pneumonia, we committed to ethical standards and got parental permission.

Convolutional Neural Network (CNN) modeling was used in the method we employed for diagnosing pneumonia. Mel-frequency Cepstral Coefficients (MFCCs), which were taken from data on respiratory sound, were used in the model. For each class ("Healthy" and "Pneumonia"), we evaluated the model's performance using a number of measures, including accuracy, precision, recall, and F1-score. In order to evaluate the model's capability for discriminating, we also performed Receiver Operating Characteristic (ROC) curve analysis. The model's performance results were analyzed, showing both its advantages and disadvantages.

The outcomes showed that the created model had significant performance and accuracy in identifying pneumonia. It demonstrated competitive recall, accuracy, and F1-scores, demonstrating its efficacy in detecting pneumonia patients. The ROC analysis showed strong discriminating capacity, which further supported the model's efficiency.

The results of this study have important significance for the early diagnosis and treatment of pneumonia in toddlers. The approach might aid medical practitioners in providing correct diagnoses by allowing autonomous detection using a self-assembled digital stethoscope. For toddler's patients, this technology may result in faster treatments and better results.

In conclusion, this project advances the area of autonomous digital stethoscope-based pneumonia identification in toddlers. The findings show the model's efficacy and potential influence on enhancing young patients' access to healthcare. Toddler pneumonia may be detected early and treated quickly with further development, validation, and integration of the model into clinical procedures.

6.2 Scope of Further Development

Micro electromechanical systems are microscopic mechanical devices created utilizing microfabrication processes. MEMS devices can be utilized for a variety of tasks and typically range in size from a few micrometers to a few millimeters.

MEMS (micro-electromechanical systems) electronic stethoscope is used to hear the noises made by the human body, especially the heart and lungs. A MEMS microphone, a tiny microphone that transforms sound waves into electrical signals, is incorporated into the device. The signal is then processed and amplified to enhance the sound. Typically, silicon, glass, or polymers are used to create MEMS devices, which are then assembled utilizing processes including photolithography, etching, and deposition. These components, which can be utilized as sensors, actuators, switches, and resonators as well as other things, are frequently found in consumer electronics, medical equipment, and industrial automation systems.

6.3 MULTIPLE DISEASES DIAGNOSIS

Globally, rheumatic heart disorders (RHDs) significantly increase morbidity and mortality. Timely secondary prophylaxis initiation is crucial to lowering the burden of RHD. The future of this work is to estimate the prevalence of subclinical RHD and build a deep learning (DL) system to predict subclinical RHD using waveform data from the digital stethoscope (DS). Using a specific study ID, the PCG and ECG data can be collected from the Digital Stethoscope and will be posted via a smartphone-based app.[34]

Budget Table:

Name	Price (TK)	Quant ity	Total Price (TK)
Raspberry Pi 4 Computer Complete Set	25000	1	25,000
RODE Lavalier GO Professional- grade Wearable Microphone	8000	1	8,000
Raspberry Pi HDMI Touch Display 5 Inch	5000	1	5,000
3.7V 18650 Rechargeable Battery 2600mAH True Capacity	380	2	760
Battery Charger	600	1	600
Stethoscope	8000	1	8,000
Wire	500	1	500
Data collection	150	20	3,000
Tot al			49,860 TK

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Appendix

```
import numpy as np
import pandas as pd
import os
for dirname, _, filenames in
os.walk('C:\\Users\\sowra\\OneDrive\\Documents\\archive\\Respiratory_Sound_Database\\Respiratory_Sound_Database\\'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

#IMPORT LIBRARY
from datetime import datetime
from os import listdir
from os.path import isfile, join

import librosa
import librosa.display

import numpy as np
import pandas as pd

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout, Conv2D, MaxPooling2D, GlobalAveragePooling2D
from keras.utils import to_categorical
from tensorflow.keras.callbacks import ModelCheckpoint

from sklearn.metrics import confusion_matrix, classification_report, roc_curve, auc
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder

import matplotlib.pyplot as plt
import seaborn as sns
from matplotlib.pyplot import savefig
mypath = "C:\\Users\\sowra\\OneDrive\\Documents\\archive\\Respiratory_Sound_Database\\audio_and_txt_files\\"
filenames = [f for f in listdir(mypath) if (isfile(join(mypath, f)) and f.endswith('.wav'))]
p_id_in_file = [] # patient IDs corresponding to each file
for name in filenames:
    p_id_in_file.append(int(name[:3]))

p_id_in_file = np.array(p_id_in_file)

#MFCC
max_pad_len = 862
```

```

def extract_features(file_name):
    """
    This function takes in the path for an audio file as a string, loads it, and returns the MFCC
    of the audio"""

    try:
        audio, sample_rate = librosa.load(file_name, res_type='kaiser_fast', duration=20)
        mfccs = librosa.feature.mfcc(y=audio, sr=sample_rate, n_mfcc=40)
        pad_width = max_pad_len - mfccs.shape[1]
        mfccs = np.pad(mfccs, pad_width=((0, 0), (0, pad_width)), mode='constant')

    except Exception as e:
        print("Error encountered while parsing file: ", file_name)
        return None

    return mfccs

filepaths = [join(mypath, f) for f in filenames]
p_diag =
pd.read_csv("C:\\Users\\sowra\\OneDrive\\Documents\\archive\\Respiratory_Sound_Database\\Respirator
y_Sound_Database\\patient_diagnosis.csv", header=None)
labels1 = np.array([p_diag[p_diag[0] == x][1].values[0] for x in p_id_in_file])
features = []
for file_name in filepaths:
    data = extract_features(file_name)
    features.append(data)

print('Finished feature extraction from ', len(features), ' files')
features = np.array(features)

# PLOTTING MFCC
plt.figure(figsize=(10, 4))
librosa.display.specshow(features[7], x_axis='time')
plt.xlabel("Time(seconds)", fontsize=18)
plt.rc('xtick', labelsize=13)
plt.colorbar()
plt.title('MFCC', fontsize=20)
plt.rc('ytick', labelsize=13)
plt.tight_layout()
plt.savefig('plot1.eps', dpi=300, bbox_inches='tight')
plt.show()

features = np.array(features)
# PRINT CLASS COUNTS
unique_elements, counts_elements = np.unique(labels1, return_counts=True)
print(np.asarray((unique_elements, counts_elements)))

```

```

# PLOT CLASS COUNTS
y_pos = np.arange(len(unique_elements))
plt.figure(figsize=(12,8))
plt.bar(unique_elements, counts_elements, align='center', alpha=0.5)
plt.xticks(y_pos, unique_elements)
plt.ylabel('Count', fontsize=21)
plt.rc('ytick', labels=15)
plt.xlabel('Disease', fontsize=21)
plt.rc('xtick', labels=14.99)
plt.title('Disease Count in Sound Files', fontsize=23)
plt.savefig('plot2.eps', dpi=300, bbox_inches='tight')
plt.show()

le = LabelEncoder()
i_labels = le.fit_transform(labels1)
oh_labels = to_categorical(i_labels)
features1 = np.reshape(features1, (*features1.shape,1))

# TRAIN TEST SPLIT
x_train, x_test, y_train, y_test = train_test_split(features, oh_labels, stratify=oh_labels,
                                                    test_size=0.25, random_state = 42)

num_rows = 40
num_columns = 862
num_channels = 1

num_labels = oh_labels.shape[1]
filter_size = 2

# CNN MODEL
model = Sequential()
model.add(Conv2D(filters=16, kernel_size=filter_size,
                input_shape=(num_rows, num_columns, num_channels), activation='relu'))
model.add(MaxPooling2D(pool_size=2))
model.add(Dropout(0.2))

model.add(Conv2D(filters=32, kernel_size=filter_size, activation='relu'))
model.add(MaxPooling2D(pool_size=2))
model.add(Dropout(0.2))

model.add(Conv2D(filters=64, kernel_size=filter_size, activation='relu'))
model.add(MaxPooling2D(pool_size=2))
model.add(Dropout(0.2))

model.add(Conv2D(filters=128, kernel_size=filter_size, activation='relu'))
model.add(MaxPooling2D(pool_size=2))

```

```

model.add(Dropout(0.2))

model.add(GlobalAveragePooling2D())

model.add(Dense(num_labels, activation='softmax'))
model.compile(loss='categorical_crossentropy', metrics=['accuracy'], optimizer='adam')
#MODEL SUMMARY
model.summary()
# MODEL PRE-TRAIN ACCURACY
score = model.evaluate(x_test, y_test, verbose=1)
accuracy = 100*score[1]
print("Pre-training accuracy: %.4f%%" % accuracy)
#MODEL TRAINING
num_epochs = 135
num_batch_size = 32

callbacks = [
    ModelCheckpoint(
        filepath='mymodel2_{epoch:02d}.h5',
        # Path where to save the model
        # The two parameters below mean that we will overwrite
        # the current checkpoint if and only if
        # the `val_accuracy` score has improved.
        save_best_only=True,
        monitor='val_accuracy',
        verbose=1)
]
start = datetime.now()

history = model.fit(x_train, y_train, batch_size=num_batch_size, epochs=num_epochs,
                    validation_data=(x_test, y_test), callbacks=callbacks, verbose=1)

duration = datetime.now() - start
print("Training completed in time: ", duration)

#TRAIN AND TEST
score = model.evaluate(x_train, y_train, verbose=0)
print("Training Accuracy: ", score[1])
score = model.evaluate(x_test, y_test, verbose=0)
print("Testing Accuracy: ", score[1])

preds = model.predict(x_test)
classpreds = np.argmax(preds, axis=1)
y_testclass = np.argmax(y_test, axis=1)
n_classes=2

```

```

#ROC
fpr = dict()
tpr = dict()
roc_auc = dict()
for i in range(n_classes):
    fpr[i], tpr[i], _ = roc_curve(y_test[:, i], preds[:, i])
    roc_auc[i] = auc(fpr[i], tpr[i])
    c_names = ['Healthy', 'Pneumonia']
    #PLOT ROC
fig, ax = plt.subplots(figsize=(16, 10))
ax.plot([0, 1], [0, 1], 'k--')
ax.set_xlim([0.0, 1.0])
ax.set_ylim([0.0, 1.05])
ax.set_xlabel('False Positive Rate', fontsize=22)
plt.rc('xtick', labels=17)
ax.set_ylabel('True Positive Rate', fontsize=22)
plt.rc('ytick', labels=17)
ax.set_title('ROC Curve for Each Class', fontsize=22)
for i in range(n_classes):
    ax.plot(fpr[i], tpr[i], linewidth=3, label='ROC curve (area = %0.2f) for %s' % (roc_auc[i], c_names[i]))
ax.legend(loc="best", fontsize='20')
ax.grid(alpha=.4)
sns.despine()
plt.show()
# CLASSIFICATION REPORT
print(classification_report(y_testclass, classpreds, target_names=c_names))
#ACCURACY CURVE
plt.figure(figsize = (15,5))
plt.subplot(1,2,1)
plt.title('Accuracy Curve', fontsize=19)
plt.plot(history.history['accuracy'], label = 'training acc')
plt.plot(history.history['val_accuracy'], label = 'validation acc')
plt.rc('xtick', labels=16)
plt.rc('ytick', labels=16)
plt.xlabel('Epochs ',fontsize=16)
plt.ylabel('Accuracy',fontsize=16)
plt.legend(fontsize=12)
#LOSS CURVE

plt.plot(history.history['loss'], label = 'training loss')
plt.plot(history.history['val_loss'], label = 'validation loss')
plt.xlabel('Epochs ',fontsize=16)
plt.ylabel('Loss',fontsize=16)
plt.legend(fontsize=12)
plt.rc('xtick', labels=15)
plt.rc('ytick', labels=15)
plt.title('Loss Curve', fontsize=19)

```

লক্ষ্য: শিশুদের থেকে স্বাস্থ্যব্দের শব্দ সংগ্রহ।

তদন্তকারীর নাম:

ভূমিকা: আপনার সন্তানকে একটি গবেষণায় অংশগ্রহণের জন্য আমন্ত্রণ জানানো হচ্ছে। এই ফর্মটি আপনাকে গবেষণা সম্পর্কে সম্পূর্ণভাবে জানাতে এবং ফুসফুসের শব্দ সংগ্রহের পদ্ধতির সম্পর্কে জানান হবে। এই নথিটি মনোযোগ সহকারে পড়ার জন্য সময় নিন এবং আপনার যেকোনো প্রশ্ন আমাদেরকে জিজ্ঞাসা করুন। এই গবেষণায় অংশগ্রহণ করা বা না করার সিদ্ধান্ত আপনার সন্তানের চিকিৎসা বা স্বাস্থ্যসেবা প্রদানকারীর সাথে সম্পর্ককে প্রভাবিত করবে না।

অধ্যয়নের উদ্দেশ্য: এই অধ্যয়নের উদ্দেশ্য হল একটি স্বয়ংক্রিয় ডিজিটাল স্টেথোস্কোপ ব্যবহার করে শিশুদের থেকে ফুসফুসের শব্দ সংগ্রহ করা। ফুসফুসের শব্দগুলি হৃৎপিণ্ডের আওয়াজ এবং অন্যান্য পরিবেশগত শব্দগুলি থেকে বিচ্ছিন্ন করা হবে যাতে শব্দগুলি সুনির্দিষ্ট এবং উচ্চমানের হয়। প্রকল্পের বর্ধিত উদ্দেশ্য হল একটি মেশিন লার্নিং অ্যালগরিদম তৈরি করা যা সুস্থ এবং অস্বাস্থ্যকর ফুসফুসের শব্দের মধ্যে পার্থক্য করতে পারে। আশা করা হচ্ছে যে ডিভাইসটি স্বাস্থ্যব্দের শব্দ চিনতে সক্ষম হবে এবং শব্দটি স্বাভাবিক বা অস্বাভাবিক কিনা তা ভবিষ্যদ্বাণী করতে সক্ষম হবে। ডিজাইন করা ডিভাইসটি ক্লিনিকাল পরিবেশে সঠিকভাবে ডেটা রেকর্ড করতে পারে।

পদ্ধতি: আপনি যদি আপনার সন্তানকে এই গবেষণায় অংশগ্রহণ করতে সম্মত হন, তাহলে নিম্নলিখিত পদ্ধতিগুলি ঘটবে:

- গবেষণা দলের একজন সদস্য আপনাকে অধ্যয়নটি ব্যাখ্যা করবেন এবং আপনার যেকোনো প্রশ্নের উত্তর দেবেন।
- আপনি অংশগ্রহণ করতে সম্মত হলে, আপনাকে একটি সম্মতি ফর্ম স্বাক্ষর করতে বলা হবে।
- একটি ডিজিটাল স্টেথোস্কোপ ব্যবহার করে আপনার সন্তানের ফুসফুসের শব্দ রেকর্ড করা হবে।
- ডিজিটাল স্টেথোস্কোপ শিশুর বুকের সামনের দিক থেকে, বুকের পিছনে এবং বুকের পার্শ্বীয় দিক থেকে শব্দ সংগ্রহ করা হবে।
- প্রক্রিয়াটির জন্য মোট সময় নেওয়া হবে প্রায় ২ থেকে ৩ মিনিট।

ঝুঁকি: এই গবেষণায় অংশগ্রহণের সাথে সম্পর্কিত কোন ঝুঁকি নেই।

উপকারিতা: এই গবেষণায় অংশগ্রহণের সুবিধা হল যে সংগৃহীত ডেটা একটি ভবিষ্যদ্বাণীমূলক বিশ্লেষণ অ্যালগরিদম বিকাশে অবদান রাখতে পারে যা সুস্থ এবং অস্বাস্থ্যকর ফুসফুসের শব্দের মধ্যে পার্থক্য করতে পারে। এতে ভবিষ্যতে নিউমোনিয়া আক্রান্তদের সহজেই নির্ণয় করা যাবে।

গোপনীয়তা: সংগৃহীত তথ্য গোপন রাখা হবে এবং শুধুমাত্র গবেষণা দলের সদস্যদের দ্বারা ব্যবহার করা হবে। ডেটা একটি সুরক্ষিত সার্ভারে সংরক্ষণ করা হবে এবং অধ্যয়ন শেষ হওয়ার পরে ধ্বংস করা হবে।

যোগাযোগের তথ্য: এই গবেষণা সম্পর্কে আপনার যদি কোনো প্রশ্ন বা উদ্বেগ থাকে, তাহলে অনুগ্রহ করে প্রধান তদন্তকারীর সাথে যোগাযোগ করুন :

- সৌরভ চৌধুরী (০১৯৫৫০৬৪৬৯৪)

নীচে স্বাক্ষর করার মাধ্যমে, আপনি স্বীকার করেন যে আপনি এই সম্মতি ফর্মে দেওয়া তথ্য পড়েছেন এবং বুঝেছেন। আপনি স্বাধীনভাবে এবং স্বেচ্ছায় আপনার সন্তানকে এই গবেষণায় অংশগ্রহণ করতে সম্মতি দিচ্ছেন।

আপনি সম্মতি ফর্মে স্বাক্ষর করার পরে যে কোনও সময় ডেটা সংগ্রহ প্রক্রিয়া থেকে প্রত্যাহার করতে পারবেন।

নিবন্ধনের তারিখ

/ /

ব্যক্তিগত তথ্য

গবেষণায় ব্যক্তিগত তথ্য গোপন রাখা হবে

মায়ের নাম

বাবার নাম

শিশুর নাম

জন্ম তারিখ

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ছেলে ☐ / মেয়ে ☐

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জেলা

এলাকা

নিবন্ধনের জন্য ধন্যবাদ

স্বাক্ষর